

Humouring the Electron: Power & Power Management in Semiconductor Designs

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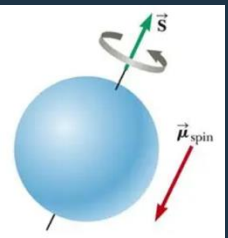
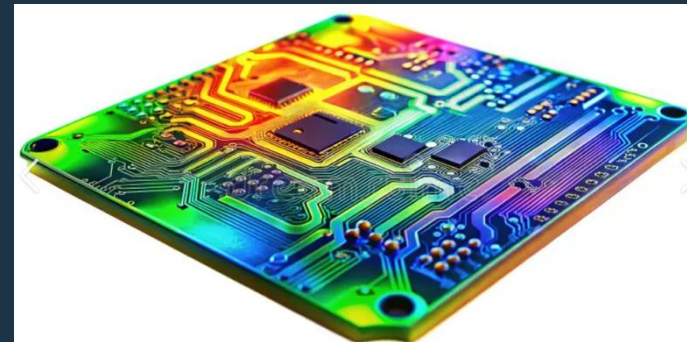
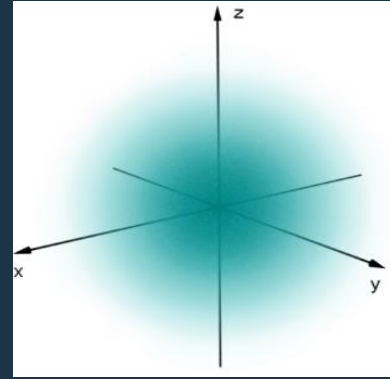
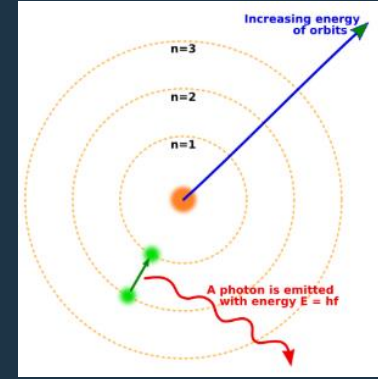


Agenda

- Electron in circuits
- Power & implications
- Power-management techniques
- Challenges with power management
- Summary

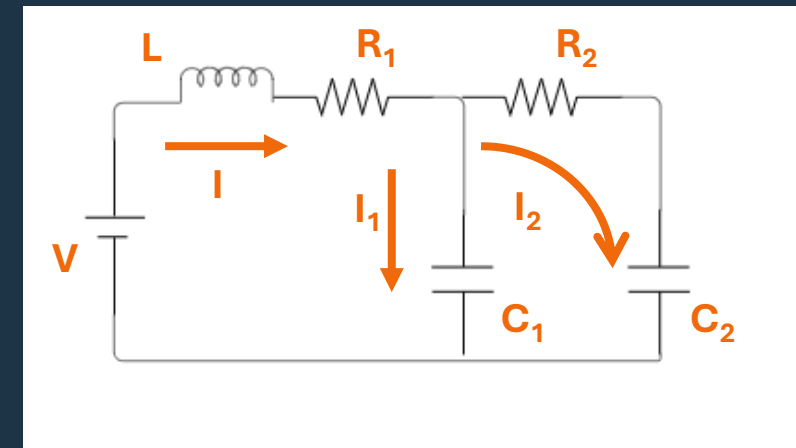
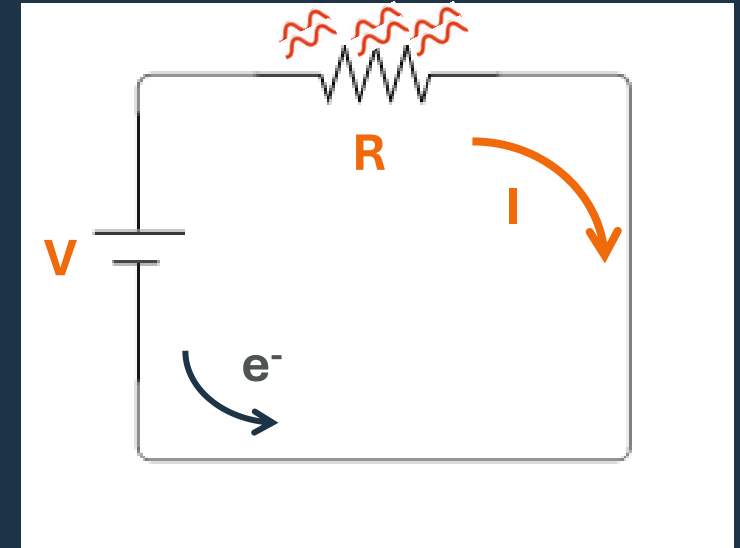
Why the Electron “Spin”?

- **electron (e^-)**
 - A sub-atomic elementary particle
 - Determines chemical, thermal and electrical properties of materials
- Observed in action (as static electricity) for several centuries, theorised in early 1800s, discovered in 1897
- Free electron movement forms the basis of electricity and electronics (electron-hole pair)
 - Often with interesting consequences

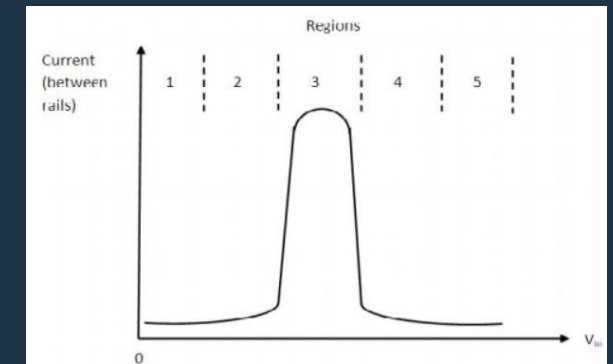
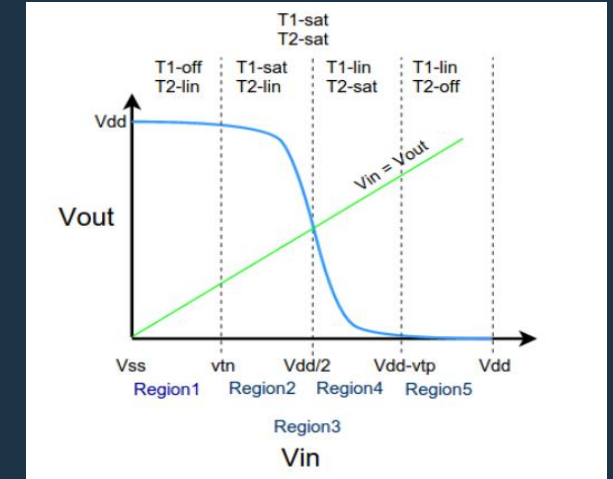
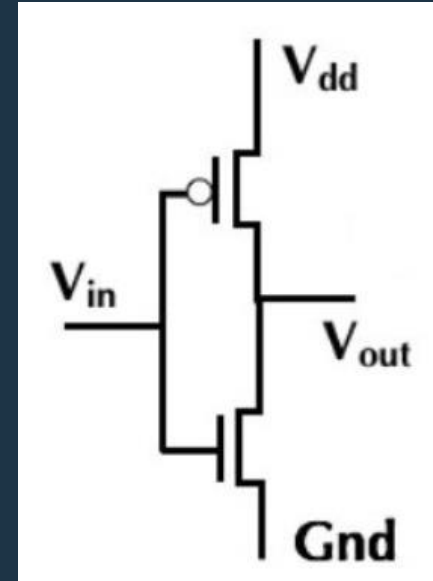
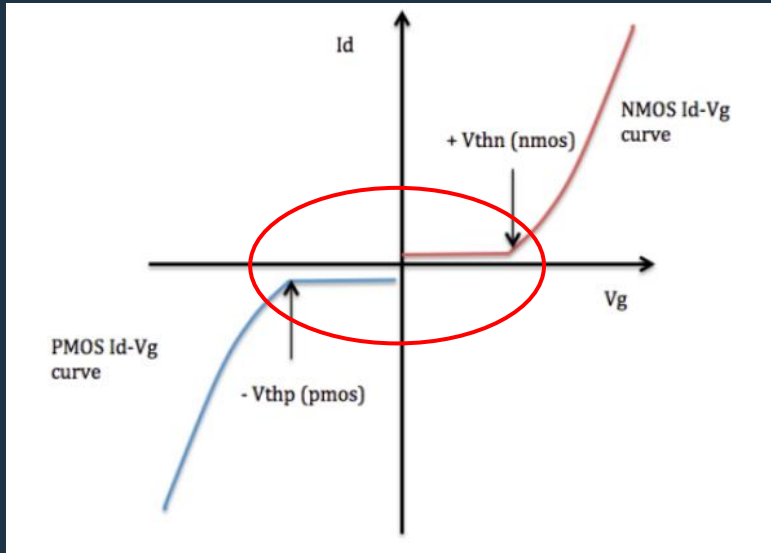
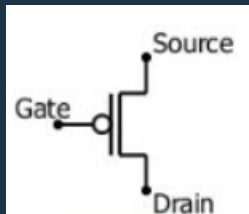
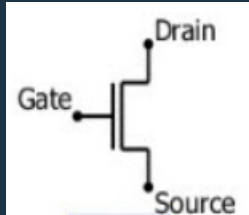


Electron in Motion

- Current = Movement of electrons
- Ohm's law: $V = I * R$ (1827)
- Joule's law: $\text{Power} = V * I = I^2 * R$ (1840)
 - Mostly heat
 - Occasionally, light
- Transient and frequency-dependent behaviour with capacitances and inductances
- Kirchhoff's (current) law: $I = I_1 + I_2$ (1845)
- Energy management is about reducing currents

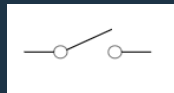


MOS Transistor Characteristics



- PMOS and NMOS IV curves show complementary but interesting trends

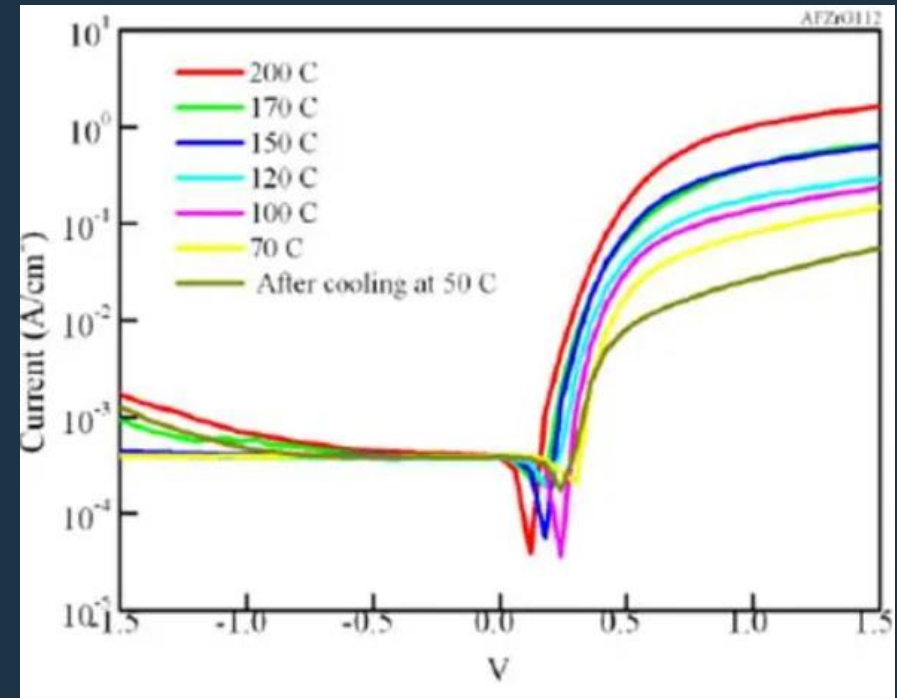
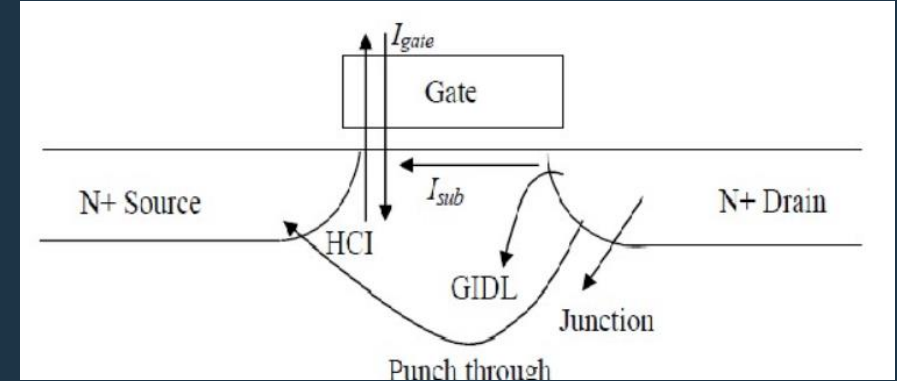
• OFF / ON states = Switch



- OFF / ON behaviour of MOS translates to currents only during transitions
 - Energy per transition $\propto C_L \cdot V^2$
 - Dynamic power dominates over leakage typically

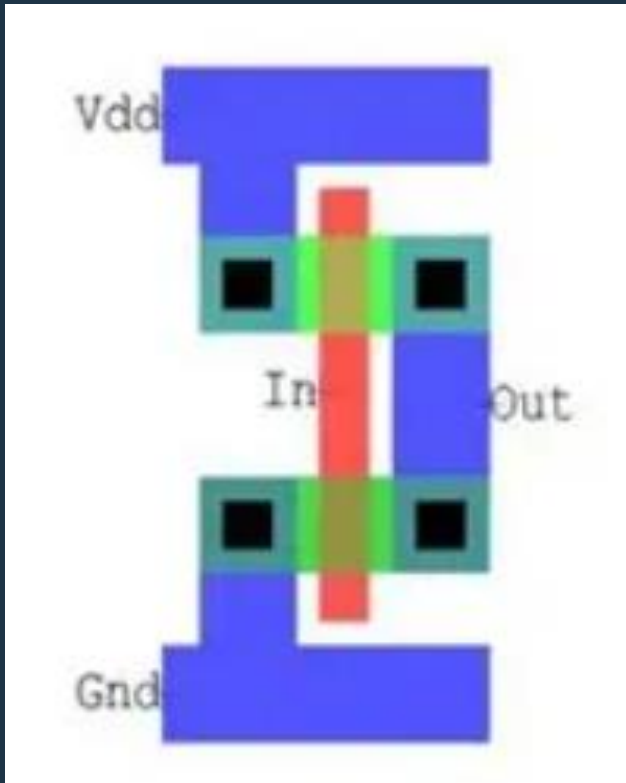
MOS Currents

- The MOS transistor is not an ideal “switch”
- Multiple “leakage” components when OFF
 - Significant variability of leakage currents with temperature
 - Nearly linear increase with voltage
- Switch has a resistance when ON
 - $R_{ON} \propto 1 / I_{ON}$
 - Increases delays in digital logic
 - Impacts core analog circuit performance



Representative trends

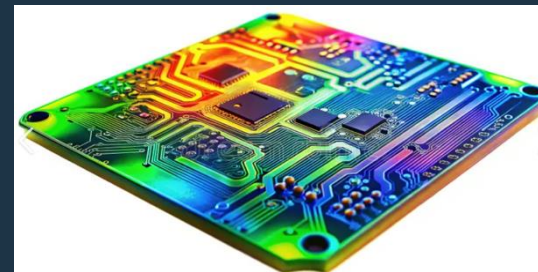
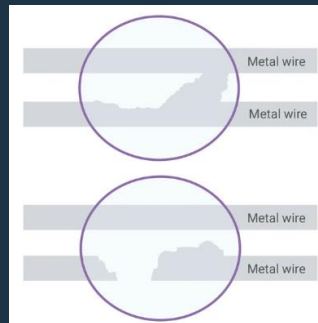
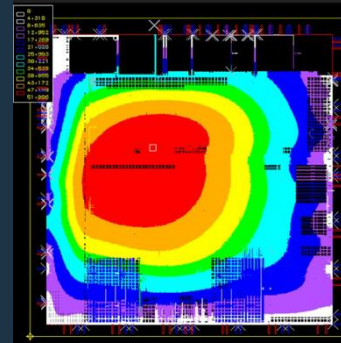
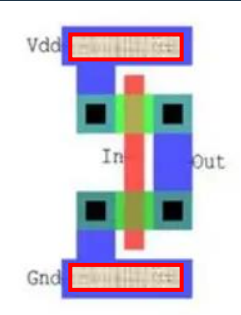
Circuit Parasitics



- Every node and geometry has unexpected parasitics
 - Resistance
 - Capacitance
 - Not much of an inductance at smaller dimensions (*usually*)
- All of these add to the circuit constituents, determining performance *and power*

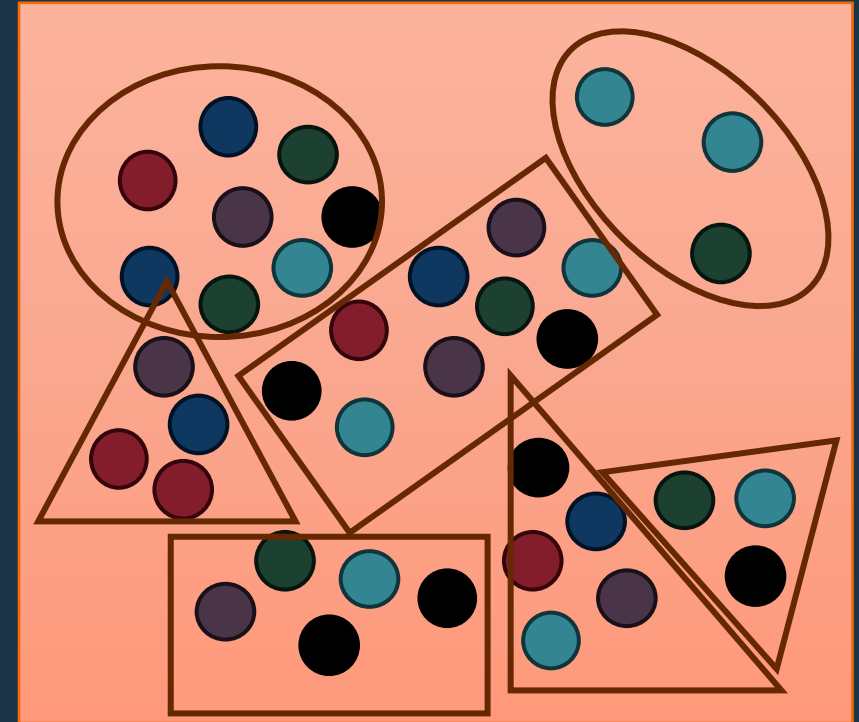
Taking Stock: Why Focus on the Electron?

- Circuit performance is directly impacted by dynamic & leakage currents
- Currents and circuit performance are impacted by parasitics
- Currents mean power dissipation → increased temperature
- Increased temperature increases currents
- Unmoderated scenarios lead to thermal run-away
- Insufficiently moderated scenarios cause other problems
 - Voltage drop (resistive drop in metals) → degraded performance
 - Electro-migration → circuit failures
 - Aggravates MOS aging & channel hot carriers → degraded performance & life-time
 - Package / system temperature issues
 - Low battery life in battery-based electronics



Power: Goodness Metrics

- Energy per transition
- Energy per operation
- Energy for a sequence of operations
- Emphasis on:
 - Optimality of operations
 - Spatial distribution of operations
 - Temporal distribution of operations

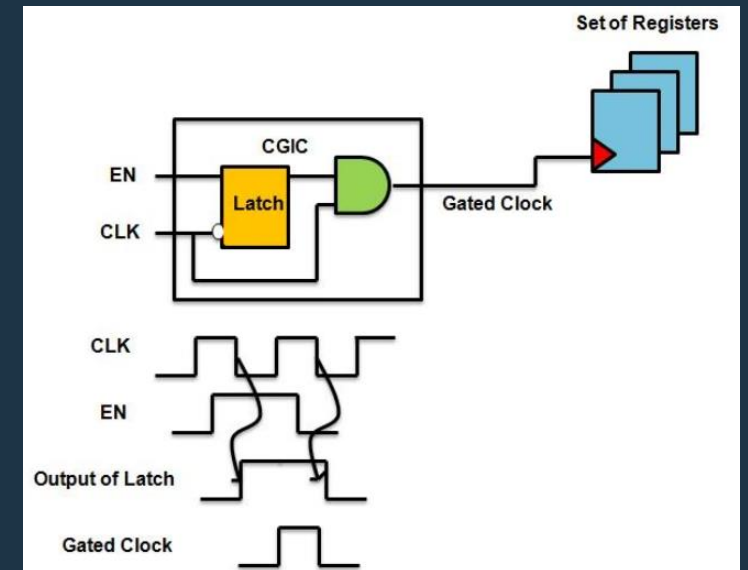
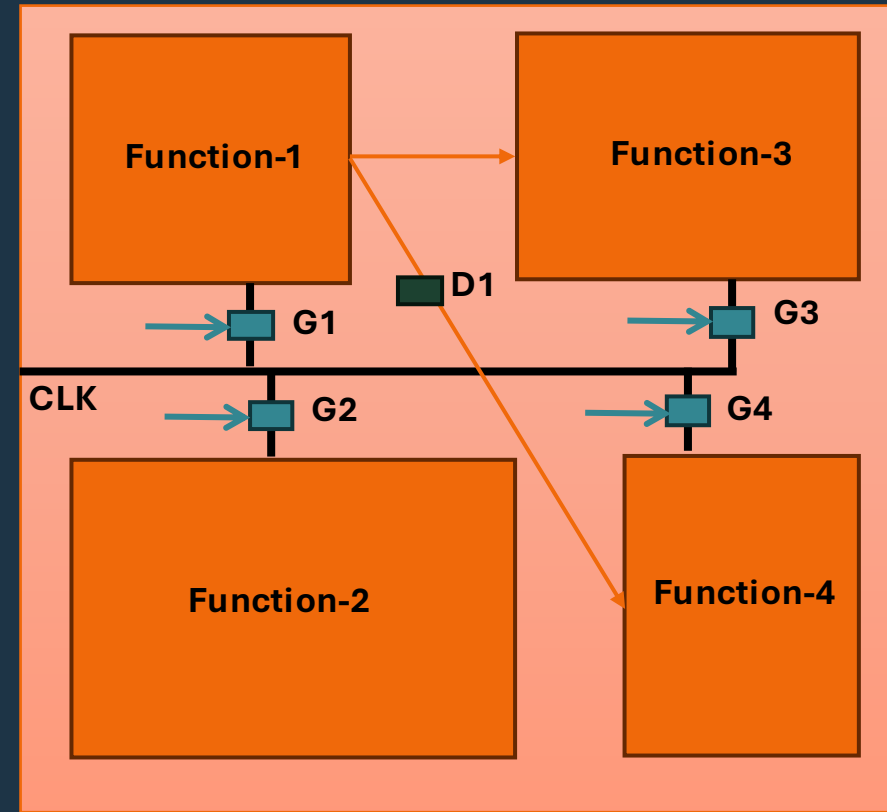


Power Management

- The art of coaxing the currents to be optimal for a given set of operations
- Several techniques
 - Toggle management – signals, clocks
 - Voltage of operation
 - Circuit refactoring
 - Turning off unwanted circuits
 - Circuit biasing
 - Architecture – multiple core vs. single core
 - Implementation choices
- Multiple challenges as well!

Toggle Management

- Clock gating
 - Turn off clocks when module not in use
- Clock frequency changes
 - Optimal clock frequency for operation → no wasted toggles
- Data gating
 - Propagate signals only to intended targets

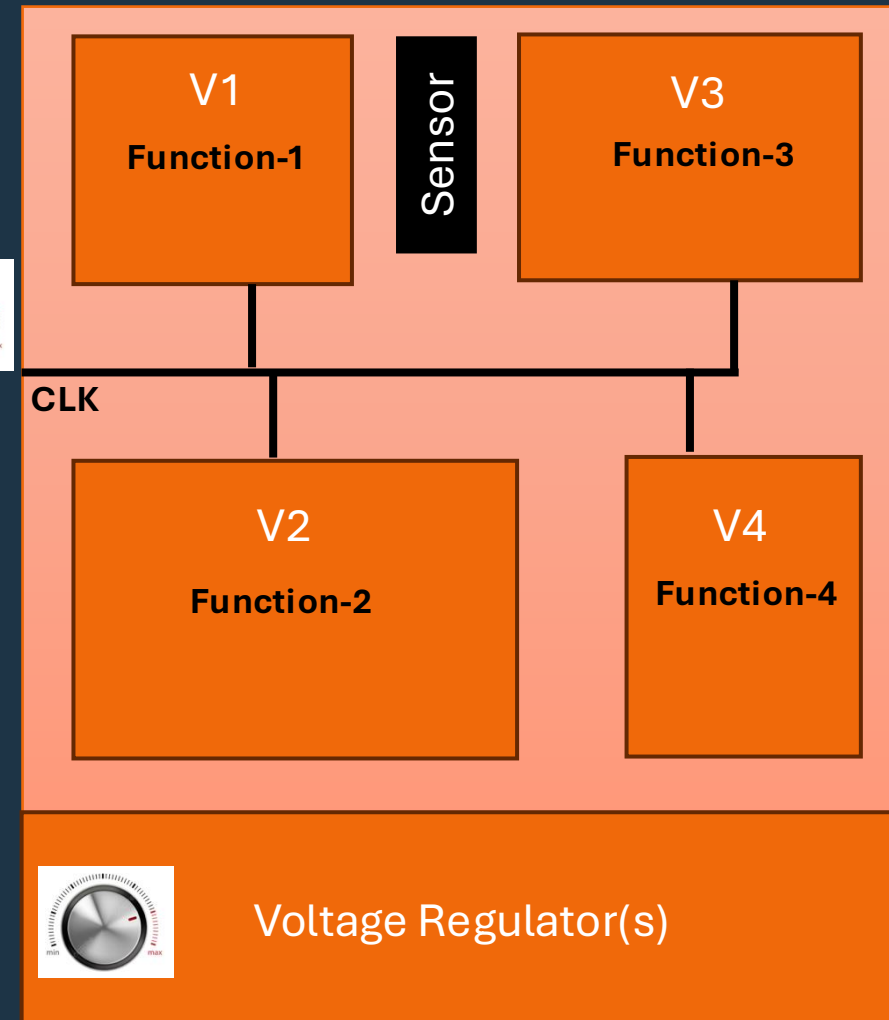
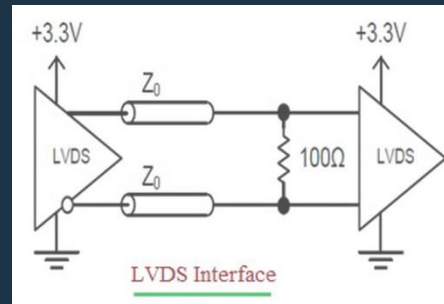


Reduced toggles = Reduced dynamic power (linear reduction)

Ascertain that currents of circuits added << savings

Voltage of Operation

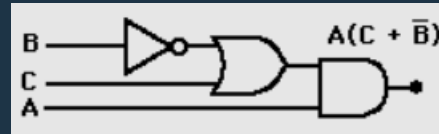
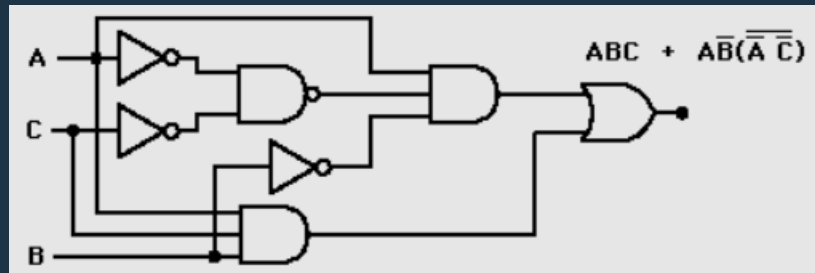
- Varying but pre-determined voltages for different functions
- Dynamic Voltage Frequency Scaling
 - Sense requirement and adjust both clock frequency & voltage dynamically
- Voltage swing reduction
 - Differential circuits for noise immunity



Reduced voltage = Reduced dynamic power (square term) + Reduced leakage (~linear)

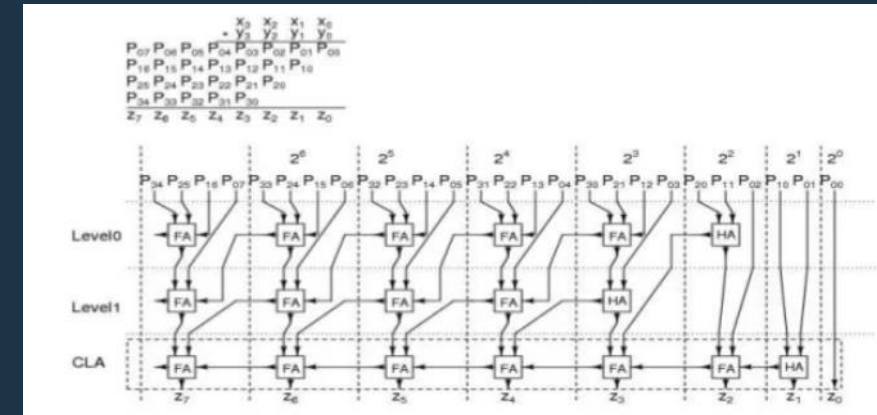
Ascertain that currents of circuits added << savings

Refactoring & Simplification

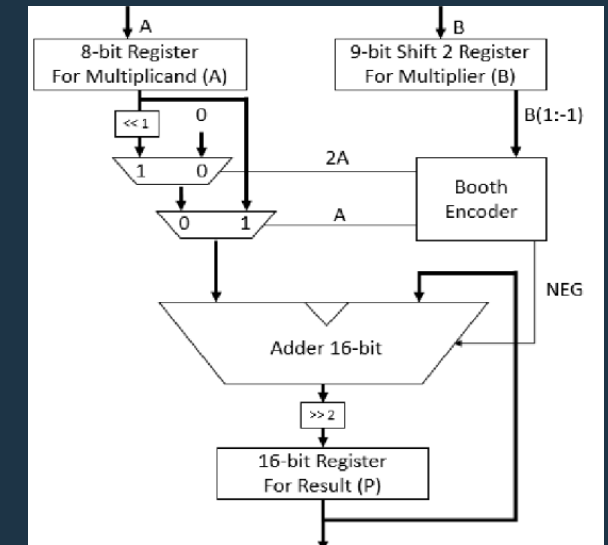


A	B	C	Out
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

- Circuit simplification can often lead to lower number of toggles
- Resource reduction → lower power
- However, it is not always that lower power circuits can be faster
- Wallace tree multiplier is fast, but is costly on power & area
- Booth multiplier is slow, but is better on power & area



Wallace Tree Multiplier

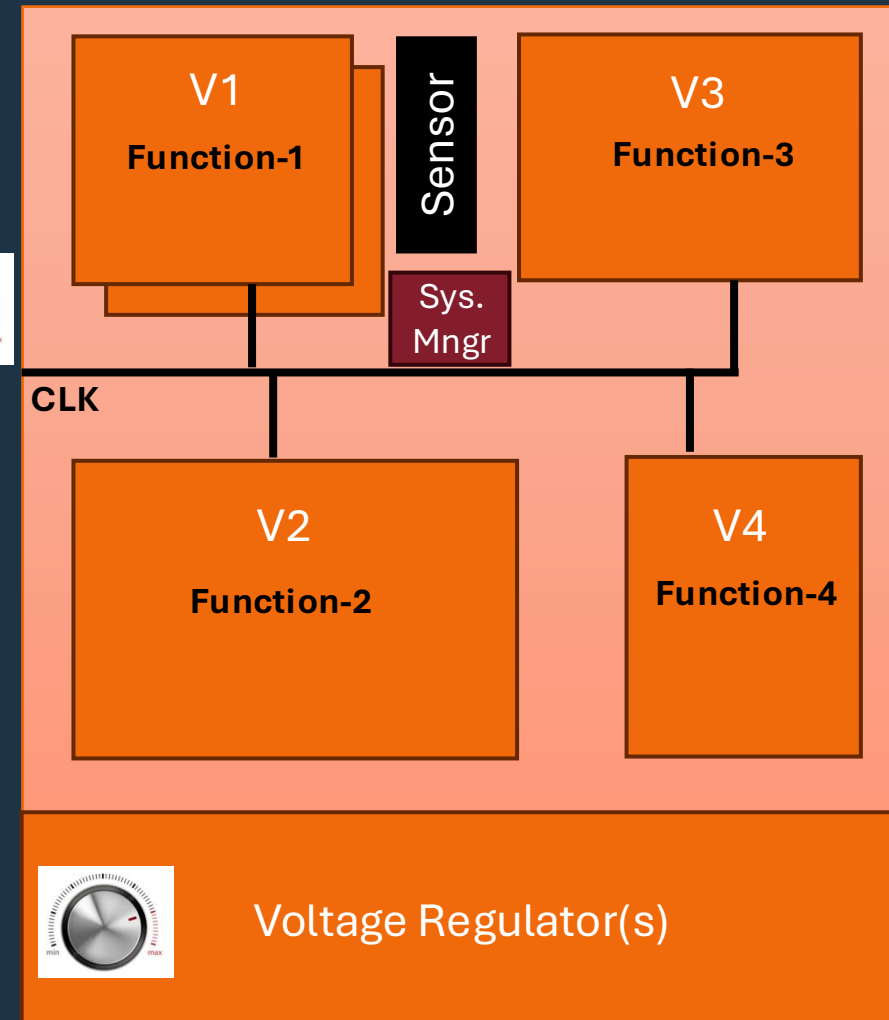


Booth Multiplier

Reduced resources = Reduced dynamic power & leakage power

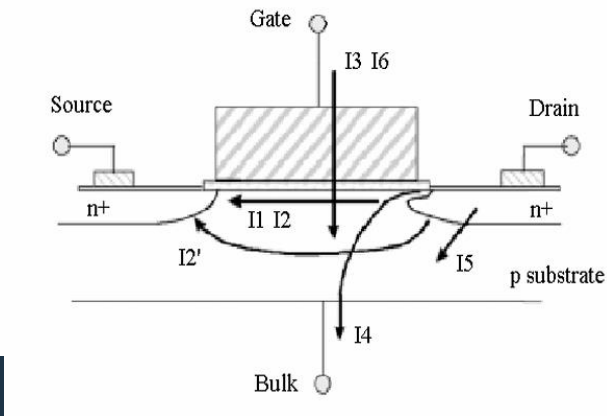
Architectural Decisions

- Need for performance → Use of additional resources
- Use of additional resources → Increased area
- Additional resources → Increased toggles (increased power)
- Increased area → Increased parasitics
- Increased parasitics → Increased power (and other problems like IR-drop)
- Are we better off having multiple “higher-level functions” operating at lower frequencies to bound overall power?
 - Single-core vs. multi-core systems
 - Performance can be individually controlled at the instance-level (voltage, clocks) with suitable architecture
 - Increased control complexity (hardware or software) to save design power

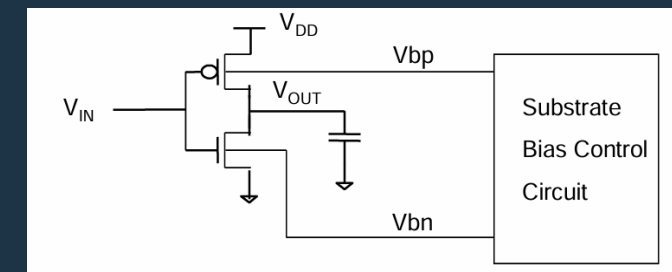
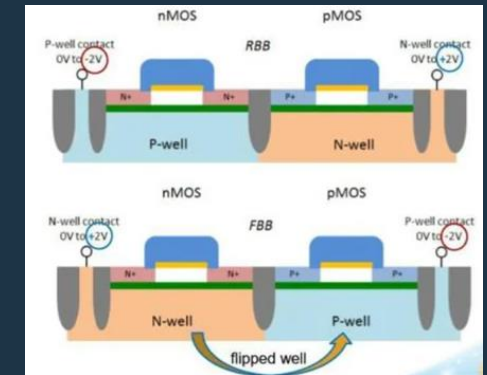
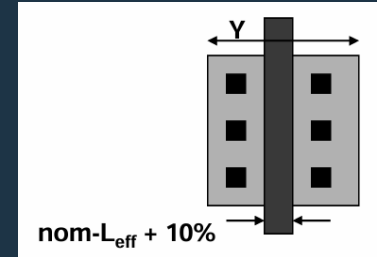
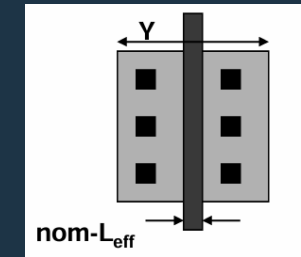


Low-leakage Design

I1: Subthreshold Current
I2: DIBL
I2': Punchthrough
I3: Thin Oxide Gate Tunneling
I4: GIDL
I5: PN Junction Current
I6: Hot Carrier Injection



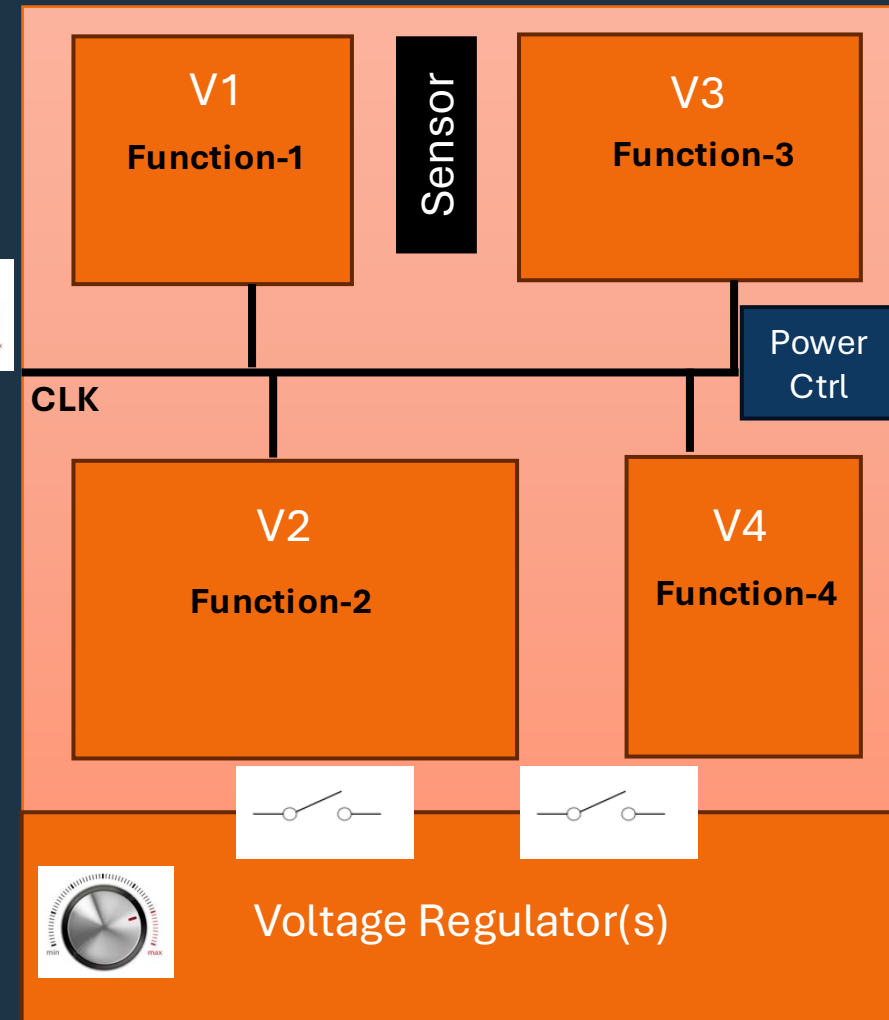
- Multiple components of leakage currents exist in MOS devices
- Increasing L reduces leakage (normally)
 - DIBL component reduced
 - Often used to slow down circuits not on critical timing paths
 - But, GIDL may go up: circuits must be sized correctly!
- Sub-threshold leakage can be managed with well-biasing → Variable threshold CMOS
 - Reverse Body Bias (RBB): increase effective V_t → lower leakage, lower performance
 - Forward Body Bias (FBB): lower effective V_t → improved performance, increased leakage



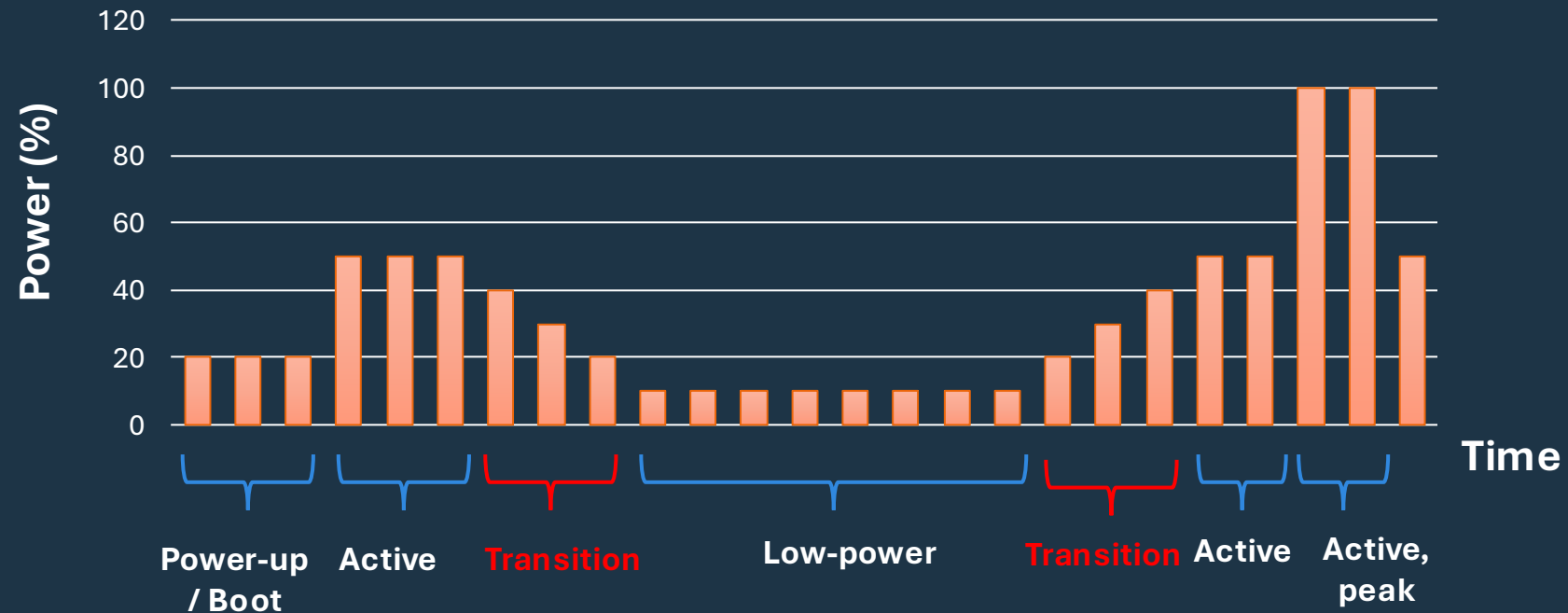
Use appropriate leakage-management techniques

Power Shut-off

- A “simple” method to save power
 - $V_x = 0$
- Can be implemented with a MOS switch
- Switch can be PMOS or NMOS
 - Determined by R_{ON} requirements
 - Determined by sizing constraints
 - And, the leakage in the switches themselves!
- A power controller in the design can control the switches
 - Automatically in hardware
 - Controlled by software



System Power States

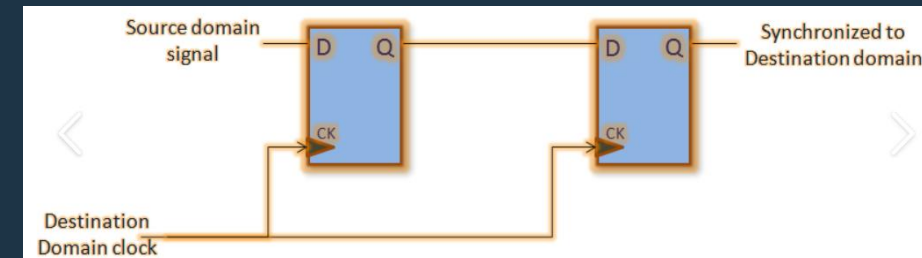
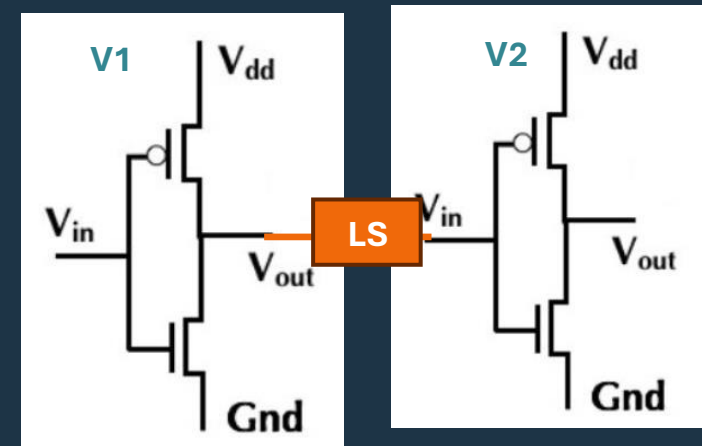
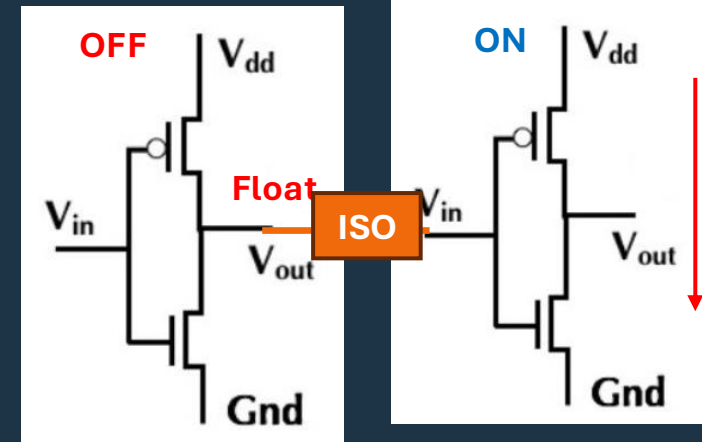


- Power expended in low-power states a critical goodness metric
- Energy spent in transitions = wasted energy
- Time spent in transition reduce the overall effectiveness of power management

Good power management solutions reduce overall energy

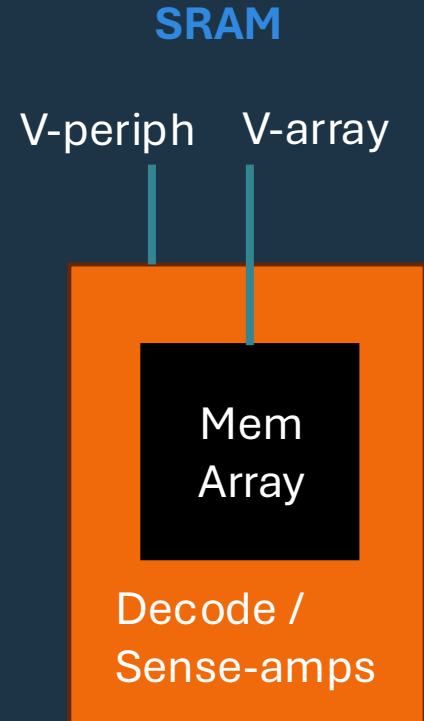
Design Care-about

- Handling power domains
 - Isolation cells are needed to avoid unintended functional behaviour and additional leakage paths
- Handling voltage domains
 - Level shifters are needed to handle electrical robustness
- Handling functional clock domains
 - Synchronisation needed to handle domain transitions
 - Focus on clocks, data transactions and resets

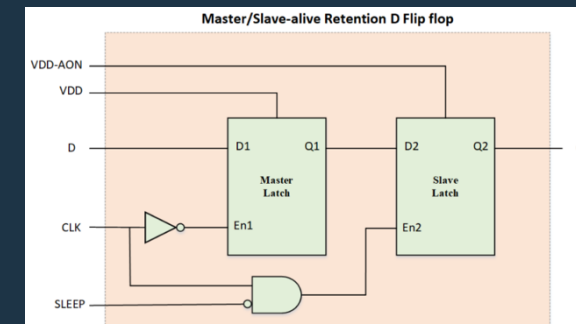


Design Care-abouts

- SW – HW interplay
 - Hardware based control of power management possible, and this avoids latencies that software control would cause
 - Better overall power
 - But, configuring the system transitions in a fixed fashion, for unknown compute work-loads, may end up being sub-optimal
 - Software based configuration is desirable in some cases
- Hardware hooks to reduce software overheads desirable
 - E.g. retaining contents of SRAM, retaining configurations of functional blocks in low power states to enable fast wake-up
 - Retention support in low power states: some sections of a circuit that is otherwise powered off kept ON
 - Expend leakage to reduce transition times

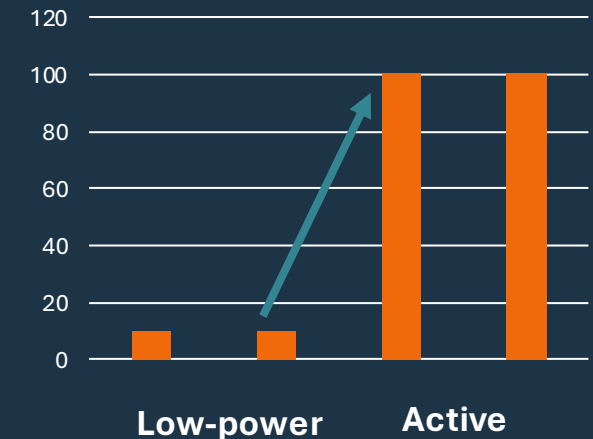
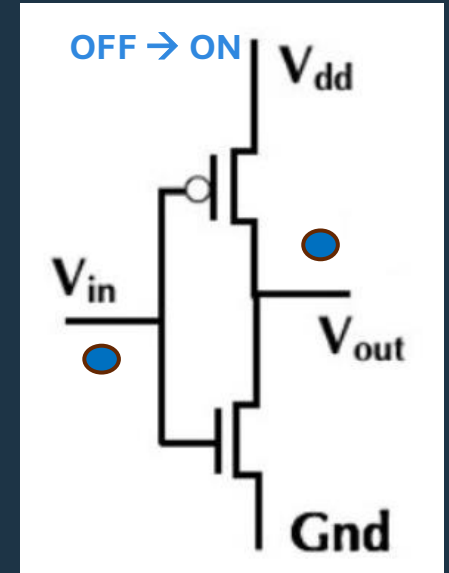


Retention Registers



Design Care-about

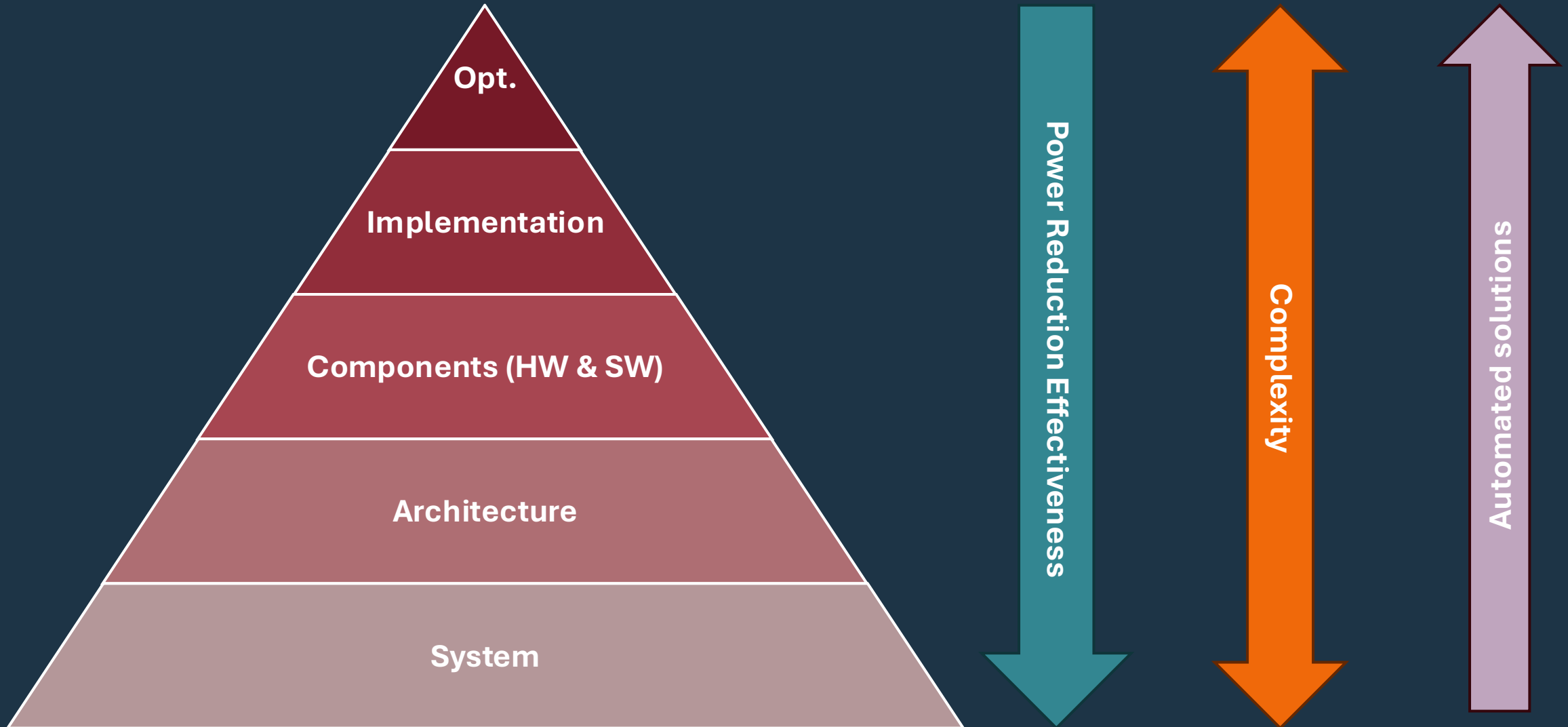
- In-rush currents whenever power is a problem to comprehend
 - All nodes in the design have to get to a quiescent state
 - Large currents → large power droops → bad functionality
 - Corruption of retained values
 - Poor start-up scenario
 - Potentially supply collapse with on-die supplies)
- Energy needed at the first clock edge is real (and is high)!



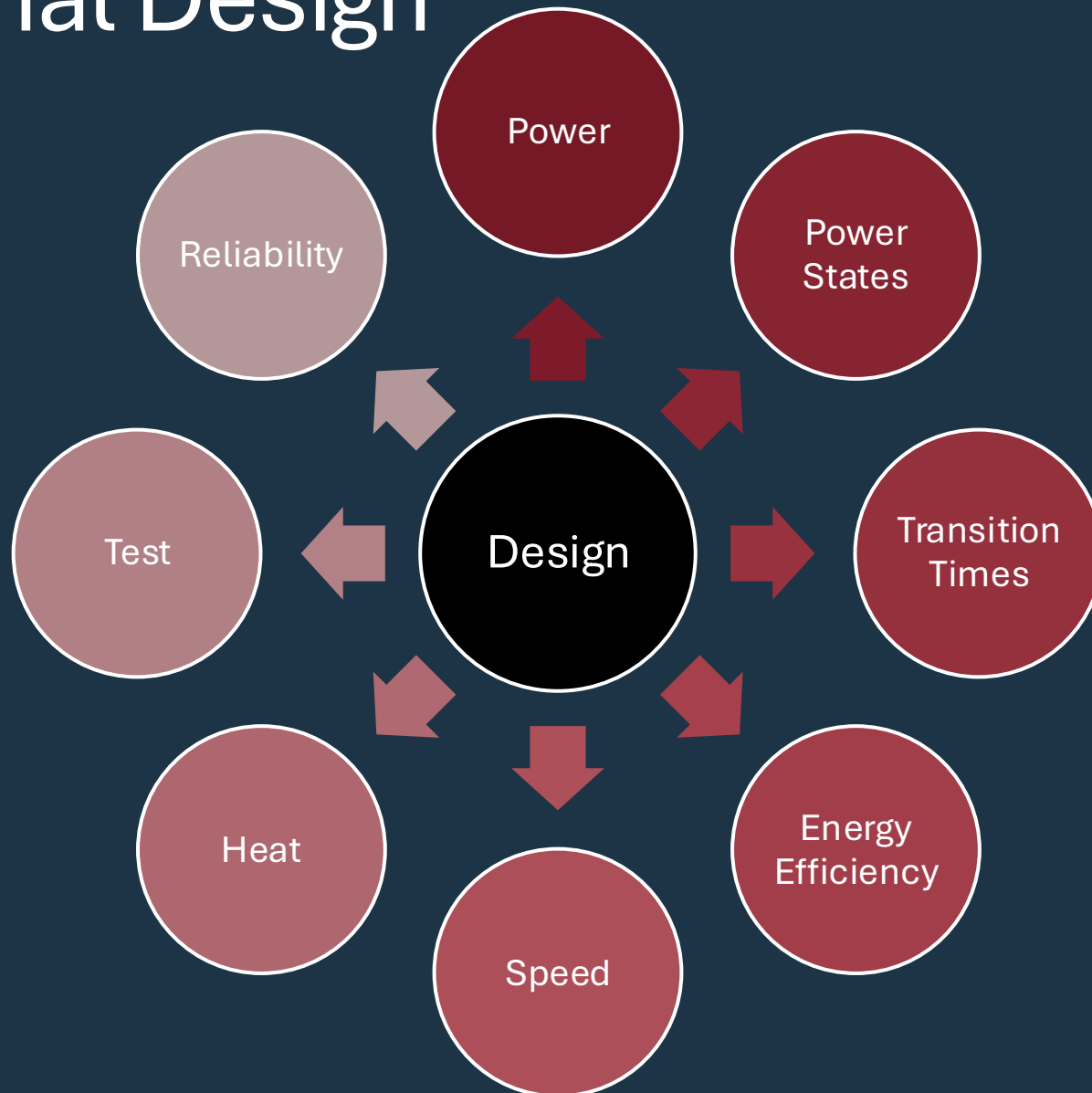
Design Care-abouts

- Inductive effects compound the problem with large current surges through power state transitions
 - Inductances can be from package pins, bond-wires or, in large die at high frequencies, even from on-die power distribution metal lines
- Multiple voltage regulators on the die are independent entities
 - Voltages may not be correlated (equal) even if the same voltage outputs are desired
 - Interactions across on-die voltage domains should be verified for uncorrelated voltage ranges
 - Often need level-shifter protection as well!

Power Reduction Efficiency



Optimal Design



An optimal power management
balances multiple goals
concurrently

Both for functionality and test

Summary & Take-aways

- Modulating electron flow in circuits is a critical focus area for modern day designs
- Designs need power management infrastructure starting from the system definition view, going all the way to the last transistor
- Power management is a multi-variable problem, with multiple optimal solutions
- EDA solutions can automate a part of the problems, but multiple challenges need design-level attention
- Several areas of “pushing the envelope” exist

Potential Research Areas

- Ultra low-power circuits – operational state, quiescent state
- Fast settling circuits – clock sources, PLLs, voltage regulators, etc.
- Configurable / scalable power/clock system
- Low-power analog signal chains & filters
- System partitioning & optimization for energy efficiency
- Work-load analysis & HW/SW partitioning
- Low-power compute infrastructure
 - Hardware accelerators
 - In-memory compute
 - Custom CPUs
- Low power test methods

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